https://sebastianl2014.github.io

# Sebastian Ljung

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## Education

**University of Cambridge, Peterhouse**

- Natural Sciences, recently graduated from 4th year integrated MSc in Astrophysics.
  MSc+11 months internship experience in astrophysics & embryology (statistical analysis).
- **Research review** (1st class), Kavli Insitute for Cosmology [4 months, part-time]: 2022
  ↳ On high-z galaxies with JWST, with Dr. Laporte (Dr. Tacchella's group). >3 years of group meetings (still going)
- **Computing project** (1st class) [4 weeks]: Vorticies in Incompressible Schrödinger Flow. 2024
- MSc: Advanced Stellar Evolution, Astro-statistics, Rel. astrophys. & cosmology (+lectures: GR, QFT)
- BSc: General relativity, Astro-fluid dynamics, Electrodynamics & Optics, Advanced QM, Stat-mech, Classical field theory & phase transitions, Particle & nuclear physics, Scattering theory.

## Publications

- Ljung S., Gilkis A., Tacchella S. (in prep)
  "Mass Transfer Accretion Disks in Differentially Rotating Binaries"
- Taubenschmid-Stowers J., Rostovskaya M., Santos F., Ljung S., ..., Reik Wolf 2022
  "Modelling human zygotic genome activation in 8C-like cells in vitro"

## Research Experience

- **MSc project supervised by Dr. Tacchella & Dr. Gilkis, Kavli Institute of Cosmology:**
  → (For Stars group): Implement mass transfer accretion disk model to enable stable modelling of differentially rotating binaries in MESA. (in prep)
  → (For Extragalactic group): Part of collaboration: Generate grid of binaries, for conversion into spectral isochrones for SED fitting by Ben Johnson (Prospector) & Dr Tacchella's group.
  - Independently formulated new analytic solution to Prof. Paczynski's (1991) numerical model on polytropic accretion disks around critically rotating main sequence stars.
  - Incorporated supernovae & transition into compact object binaries within MESA.
  - Experience with bash scripts, Fortran, and HPC cluster.
- **Internship in Prof. Anastasia Fialkov's group, Institute of Astronomy [8 weeks]:** 2023
  ↳ Modified semi-analytic A-SLOTH simulation (Fortran) to account for the influence of super-massive black holes (SMBH) on the modelled nuclear stellar clusters & Pop III black hole binaries.
  - After literature review, proposed a side-project to introduce SMBHs
  - Incorporated equations from publications and justified modelling approximations.
  - Implemented and determined impact of eccentricity oscillations on Stochastic GW Background.
- **Internship at Dr Michael Imbeault lab, Genetics Dept. Cambridge [10 weeks]:** 2022
  - Handling large datasets (>30GB), w/o cluster, in a RAM and time-efficient manner.
  - Spearheaded side-project in collaboration with Prof. Reik's lab
- **Internship at Prof. Wolf Reik lab, BI Cambridge [9 weeks]:** 2021
  ↳ 1st-order ODEs & dimensional reduction inferring cell state transitions.(analogy: galaxy phase-space flow.)
  - Co-authorship for answering review question, computationally bypassing months of costly experiments.
  - Presented results of experimental + computational (side-project) at conference poster session.